where on the earth's surface the photograph was taken, and plot this on Google Maps, showing how spatial context can be extracted from a digital image of a physical object.

Listing 2.6 Extract geographical coordinates from a photograph and display the location the photograph was taken in Google Maps.

```
from PIL import Image
from PIL.ExifTags import TAGS, GPSTAGS
import webbrowser

def main():
  exif = Image.open("./images/rockPaintings.jpg")._getexif() #1
  gps = next((dict((GPSTAGS.get(k, k), v) for k, v in val.items())
  for tag, val in exif.items() if TAGS.get(tag) == "GPSInfo"), {})

  def convert_to_degrees(value): #2
    d = float(value[0])
    m = float(value[1])
    s = float(value[2])
    return d + (m / 60.0) + (s / 3600.0)

  lat = convert_to_degrees(gps['GPSLatitude'])
  lat *= -1 if gps['GPSLatitudeRef'] == 'S' else 1

  lon = convert_to_degrees(gps['GPSLongitude'])
  lon *= -1 if gps['GPSLongitudeRef'] == 'W' else 1

  webbrowser.open(f"https://www.google.com/maps?q={lat},{lon}") #3

if __name__ == "__main__":
  main()
```

#1 Extract GPS coordinates from EXIF data in the photograph.
#2 Convert GPS coordinates to degrees in float format.
#3 Open Google Maps with a pin at the location where the photograph was taken.

2.5.2 Coordinate transformations and conversions

So why is it so important to understand the different types of spatial reference systems when capturing physical systems for digital representation? Real-world digital twin projects frequently encounter